#1. $R = 6578$ km, $m = 100$ kg.

The original CW equations.
$$\begin{cases} \ddot{\Delta x} - 2\omega_0 \dot{\Delta y} + \omega_0^2 \Delta x = F_r/m \\ \ddot{\Delta y} - 2\omega_0 \dot{\Delta x} = F_\nu/m \\ \ddot{\Delta z} + \omega_0^2 \Delta z = F_\lambda/m \end{cases}$$

a. Since $\boxed{R = 6578\ \text{km}}$, a distance from the origin of the reference frame to a location of the spacecraft has to be less than or equal to $0.03R \approx 197.34$ km.

From $m\ddot{\Delta x} = F_r$, $m\ddot{\Delta y} = F_\nu$, and $m\ddot{\Delta z} = F_\lambda$,
$$\begin{cases} \Delta x(t) = \dot{x}_0 t + x_0 \\ \Delta y(t) = \dot{y}_0 t + y_0 \\ \Delta z(t) = \dot{z}_0 t + z_0 \end{cases}$$

Those 3 equations are linear approximation of the 3 equations of the hint
Concider $x(t)$ of the hint
Nonlinear terms are $\sin(\omega t)$ and $\cos(\omega t)$, and ωt and 1 is approximated terms of $\sin \omega t$ and $\cos \omega t$.
Since required accuracy is 3%,
$$\omega t - \sin(\omega t) \le 0.03 \sin(\omega t) \ \& \ 1 - \cos(\omega t) \le 0.03 \cos(\omega t)$$
$$\Rightarrow \omega t \le 1.03 \sin(\omega t) \ \& \ 1 \le 1.03 \cos(\omega t)$$
$$\Rightarrow \omega t \le 0.4200 \quad \& \quad \omega t \le 0.2419.$$
$$\Rightarrow \omega t \le 0.2419.$$
$$\therefore t \le \frac{0.2419}{\omega} = 0.2419 \sqrt{\frac{R^3}{\mu}} \approx 204.4128 \text{ sec} \approx 3.4069 \text{ min}.$$

b. The initial state $X_0 = [-10, 0, 0, 0, 0, 0]^T$

$$\dot{X} = \begin{bmatrix} 0&1&0&0&0&0 \\ 0&0&0&0&0&0 \\ 0&0&0&1&0&0 \\ 0&0&0&0&0&0 \\ 0&0&0&0&0&1 \\ 0&0&0&0&0&0 \end{bmatrix} X_0^+ = AX_0^+ \quad \text{where } X_0^+ = \begin{bmatrix} -10 \\ \dot{x}_0^+ \\ 0 \\ \dot{y}_0^+ \\ 0 \\ \dot{z}_0^+ \end{bmatrix}$$

$$\Rightarrow X(t) = e^{At} X_0^+ = \left[\sum_{i=0}^{\infty} (At)^i\right] X_0^+ = (I + At) X_0^+$$

$$\Rightarrow \begin{bmatrix} 0 \\ \dot{x}_f \\ 0 \\ \dot{y}_f \\ 0 \\ \dot{z}_f \end{bmatrix} = \begin{bmatrix} 1&T&0&0&0&0 \\ 0&1&0&0&0&0 \\ 0&0&1&T&0&0 \\ 0&0&0&1&0&0 \\ 0&0&0&0&1&T \\ 0&0&0&0&0&1 \end{bmatrix} \begin{bmatrix} -10 \\ \dot{x}_0^+ \\ 0 \\ \dot{y}_0^+ \\ 0 \\ \dot{z}_0^+ \end{bmatrix}$$

$$\Rightarrow \begin{cases} -10 + \dot{x}_0^+ T = 0. \\ 0 + \dot{y}_0^+ T = 0 \\ 0 + \dot{z}_0^+ T = 0 \end{cases} \quad \& \quad \begin{cases} \dot{x}_0^+ = \dot{x}_f^- \\ \dot{y}_0^+ = \dot{x}_f^- \\ \dot{z}_0^+ = \dot{z}_f^- \end{cases}$$

$$\Rightarrow \dot{x}_0^+ = 1.667 \text{ km/s} = \dot{x}_f, \quad 0 = \dot{y}_0^+ = \dot{z}_0^+ = \dot{y}_f^- = \dot{z}_f^-$$

Thus $\Delta \vec{V}_0 = 1.667\,\hat{i}$ km/s and $\Delta \vec{V}_f = -1.667\,\hat{i}$ km/s

#2. EoM for the Earth-Moon circular restricted 3-body problem.

$$\begin{cases} \ddot{x} - 2\omega_0\dot{y} - \omega_0^2 x = -\dfrac{\mu_e(x-D_e)}{r_e^3} - \dfrac{\mu_m(x+D_m)}{r_m^3} + \dfrac{F_x}{m} \cdots (a) \\ \ddot{y} + 2\omega_0\dot{x} - \omega_0^2 y = -\dfrac{\mu_e y}{r_e^3} - \dfrac{\mu_m y}{r_m^3} + \dfrac{F_y}{m} \quad (b) \\ \ddot{z} = -\dfrac{\mu_e z}{r_e^3} - \dfrac{\mu_m y}{r_m^3} + \dfrac{F_z}{m} \quad (c) \end{cases}$$

a. If $L_4 : (\bar{x}, \bar{y}, \bar{z}) = (-187699.3, 333201.5, 0)$ km is a equilibrium point, 3 EoM hold for L_4 with zero velocity and zero acceleration

(a): $-\omega_0^2\bar{x} = -\dfrac{\mu_e(\bar{x}-D_e)}{r_e^3} - \dfrac{\mu_m(\bar{x}+D_m)}{r_m^3} + \cancel{\dfrac{F_x}{m}}^{\,0}$

Left side : $-\omega_0^2\bar{x} = (2.66199e\text{-}06)^2(187699.3) \approx 1.33301e\text{-}06$.

Right side : $-\dfrac{\mu_e(\bar{x}-D_e)}{r_e^3} - \dfrac{\mu_m(\bar{x}+D_m)}{r_m^3} \cdots (d)$

$$\begin{cases} r_e = \sqrt{(\bar{x}-D_e)^2 + \bar{y}^2 + \bar{z}^2} \\ \quad = \sqrt{(187699.3 - 4674)^2 + (333201.5)^2} \approx 384747.6136 \\ r_m = \sqrt{(\bar{x}+D_m)^2 + \bar{y}^2 + \bar{z}^2} \end{cases}$$

$= \sqrt{(-187699.3 + 380073)^2 + (333201.5)^2} \approx 384747.8163$.

$\Rightarrow \dfrac{-\mu_e(\bar{x}-D_e)}{r_e^3} = \dfrac{-398601(-187699.3 - 4674)}{384747.6136^3} \approx +1.3463e\text{-}06$

$-\dfrac{\mu_m(\bar{x}+D_m)}{r_m^3} = \dfrac{-4887(-187699.3 + 380073)}{384747.8163^3} \approx -1.6507e\text{-}08$

(d) : 1.3628e-06.

(b): $-\omega_0^2\bar{y} = -\dfrac{\mu_e\bar{y}}{r_e^3} - \dfrac{\mu_m\bar{y}}{r_m^3} + \cancel{\dfrac{F_y}{m}}^{\,0}$

Left side : $-\omega_0^2\bar{y} = -(2.66199e\text{-}06)^2(333201.5) \approx -2.3611e\text{-}06$.

Right side : $-\dfrac{\mu_e\bar{y}}{r_e^3} - \dfrac{\mu_m\bar{y}}{r_m^3}$

$= -\dfrac{398601 \times 333201.5}{384747.6136^3} - \dfrac{4887 \times 333201.5}{384747.8163^3}$

$\approx -2.3319e\text{-}06 - 2.8590e\text{-}08 = 2.3605e\text{-}06$.

(c) Both sides are zero.

Since (a), (b), (c) hold for L_4, then L_4 is equilibrium point.

b. Define $x = \bar{x} + \Delta x$, $y = \bar{y} + \Delta y$, and $z = \bar{z} + \Delta z = 0 + \Delta z$.

(a) Left Side : $(\ddot{\cancel{\bar{x}}} + \Delta\ddot{x}) - 2\omega_0(\dot{\cancel{\bar{y}}} + \Delta\dot{y}) - \omega_0^2(\bar{x} + \Delta x)$

$= \Delta\ddot{x} - 2\omega_0\Delta\dot{y} - \omega_0^2\bar{x} - \omega_0^2\Delta x$

Right Side $\boxed{r_e^{-3}} = \{(\bar{x} + \Delta x - D_e)^2 + (\bar{y} + \Delta y)^2 + \Delta z^2\}^{-3/2}$

$= \{(\bar{x} - D_e)^2 + \bar{y}^2 + 2\Delta x(\bar{x} - D_e) + 2\Delta y\bar{y}\}^{-3/2}$

$= \boxed{\{(\bar{x} - D_e)^2 + \bar{y}^2\}^{-3/2}}\{1 + 2\Delta x C_1 + 2\Delta y C_2\}^{-3/2}$

$= \{(\bar{x} - D_e)^2 + \bar{y}^2\}^{-3/2}\{1 - 3\Delta x C_1 - 3\Delta y C_2\}$

$$\text{where } \begin{cases} C_1 = (\bar{x}-D_e) / \{(\bar{x}-D_e)^2 + \bar{y}^2\} = (\bar{x}-D_e)/R_e \\ C_2 = \bar{y} / \{(\bar{x}-D_e)^2+\bar{y}^2\} = \bar{y}/R_e \end{cases}$$

$$r_m^{-3} = \{(\bar{x}+\Delta x + D_m)^2 + (\bar{y}+\Delta y)^2 + \Delta z^2\}^{-3/2}$$
$$= \{(\bar{x}+D_m)^2 + \bar{y}^2 + 2\Delta x(\bar{x}+D_m) + 2\Delta y \bar{y}\}^{-3/2}$$
$$= \{(\bar{x}+D_m)^2 + \bar{y}^2\}^{-3/2}\{1 + 2\Delta x C_3 + 2\Delta y C_4\}^{-3/2}$$
$$= \{(\bar{x}+D_m)^2+\bar{y}^2\}^{-3/2}\{1 - 3C_3\Delta x - 3C_4\Delta y\}$$

$$\text{where } \begin{cases} C_3 = (\bar{x}+D_m) / \{(\bar{x}+D_m)^2 + \bar{y}^2\} = (\bar{x}+D_m)/R_m \\ C_4 = \bar{y} / \{(\bar{x}+D_m)^2+\bar{y}^2\} = \bar{y}/R_m \end{cases}$$

$$\Rightarrow -\frac{\mu_e(\bar{x}+\Delta x - D_e)}{r_e^3} - \frac{\mu_m(\bar{x}+\Delta x+D_m)}{r_m^3}$$
$$= -\frac{\mu_e}{R_e^3}(\bar{x}-D_e+\Delta x)(1-3\Delta x C_1 - 3\Delta y C_2) - \frac{\mu_m}{R_m^3}(\bar{x}+D_m+\Delta x)(1-3\Delta x C_3 - 3\Delta y C_4)$$
$$= \cancel{-\frac{\mu_e}{R_e^3}(\bar{x}-D_e)} \cancel{-\frac{\mu_m}{R_m^3}(\bar{x}+D_m)} - \frac{\mu_e}{R_e^3}\Delta x - \frac{\mu_m}{R_m^3}\Delta x$$
$$+3\frac{\mu_e}{R_e^3}(\bar{x}-D_e)\left(\frac{\bar{x}-D_e}{R_e}\Delta x + \frac{\bar{y}}{R_e}\Delta y\right) + 3\frac{\mu_m}{R_m^3}(\bar{x}+D_m)\left(\frac{\bar{x}+D_m}{R_m}\Delta x + \frac{\bar{y}}{R_m}\Delta y\right)$$

$$\Rightarrow \ddot{\Delta x} - 2\omega_0\dot{\Delta y} - \omega_0^2\Delta x = \frac{\mu_e}{R_e^3}\Delta x\left(\frac{3(\bar{x}-D_e)^2}{R_e} - 1\right) + \frac{\mu_m}{R_m^3}\Delta x\left(\frac{3(\bar{x}+D_m)^2}{R_m} - 1\right)$$
$$+ 3\left(\frac{\mu_e}{R_e^4}\bar{y}(\bar{x}-D_e) + \frac{\mu_m}{R_m^4}\bar{y}(\bar{x}+D_m)\right)\Delta y + \frac{F_x}{m}.$$

$$\Rightarrow \ddot{\Delta x} - 2\omega_0\dot{\Delta y} - \omega_0^2\Delta x = (284991.0000 + 3162.7633)\omega_0^2\Delta x$$
$$- 487570\,\omega_0^2\Delta y + F_x/m.$$
$$= 288153.7633\,\omega_0^2\Delta x - 487570\,\omega_0^2\Delta y + F_x/m$$

$$\therefore \ddot{\Delta x} - 2\omega_0\dot{\Delta y} - 288152.7633\,\omega_0^2\Delta x + 487570\,\omega_0^2\Delta y = F_x/m.$$

(b) Left Side : $\ddot{\Delta y} + 2\omega_0\dot{\Delta x} - \cancel{\omega_0^2\bar{y}} - \omega_0^2\Delta y.$

Right Side : $-\frac{\mu_e}{R_e^3}(\bar{y}+\Delta y)(1-3\Delta x C_1 - 3C_2\Delta y) - \frac{\mu_m}{R_m^3}(\bar{y}+\Delta y)(1-3C_3\Delta x - 3C_4\Delta y)$

$$= \cancel{-\frac{\mu_e}{R_e^3}\bar{y}} \cancel{-\frac{\mu_m}{R_m^3}\bar{y}} - \frac{\mu_e}{R_e^3}\Delta y - \frac{\mu_m}{R_m^3}\Delta y + 3\bar{y}\left(\frac{\bar{x}+D_m}{R_m}\Delta x + \frac{\bar{y}}{R_m}\Delta y\right)\frac{\mu_m}{R_m^3}$$
$$+ 3\bar{y}\left(\frac{\bar{x}-D_e}{R_e}\Delta x + \frac{\bar{y}}{R_e}\Delta y\right)\frac{\mu_e}{R_e^3}$$

$$\Rightarrow \ddot{\Delta y} + 2\omega_0\dot{\Delta x} - \omega_0^2\Delta y = 3\bar{y}\Delta x\left(\frac{\mu_e(\bar{x}-D_e)}{R_e^4} + \frac{\mu_m(\bar{x}+D_m)}{R_m^4}\right)$$
$$+ \Delta y\left(\frac{3\bar{y}^2}{R_e} - 1\right)\frac{\mu_e}{R_e^3} + \Delta y\left(\frac{3\bar{y}^2}{R_m} - 1\right)\frac{\mu_m}{R_m^3} + \frac{F_y}{m}.$$
$$= -487569.6068\,\omega_0^2\Delta x + F_y/m.$$
$$+(854981.0479 + 10482.37021)\omega_0^2\Delta y$$

$$\therefore \ddot{\Delta y} + 2\omega_0\dot{\Delta x} + 487569.6068\,\omega_0^2\Delta x - 865464.4181\,\Delta y = F_y/m.$$

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(c) $-\frac{\mu_e \Delta z}{r_e^3} - \frac{\mu_m \Delta z}{r_m^3} = \frac{-\mu_e}{R_e^3}\Delta z(1-3C_1\Delta x-3C_2\Delta y) - \frac{\mu_m}{R_m^3}\Delta z(1-3C_3\Delta x-3C_4$

$$= -\frac{\mu_e}{R_e^3}\Delta z - \frac{\mu_m}{R_m^3}\Delta z$$

$$\Rightarrow \Delta\ddot{z} = -\left(\frac{\mu_e}{R_e^3} + \frac{\mu_m}{R_m^3}\right)\Delta z + \frac{F_z}{m}$$

$$\Rightarrow \Delta\ddot{z} + 1.000\,\omega_0^2 \Delta z = F_z/m.$$

c. $\begin{bmatrix} \Delta\dot{z} \\ \Delta\ddot{z} \end{bmatrix} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\begin{bmatrix} \Delta z \\ \Delta\dot{z} \end{bmatrix} \Rightarrow \left|\lambda I - \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}\right| = \begin{vmatrix} \lambda & -1 \\ -1 & \lambda \end{vmatrix} = \lambda^2 - 1 = 0.$

Thus the eigenvalues of out of plane are $\pm j\omega_0$.

d. Since the eigenvalues of out of plane are pure complex number, then the linearized EoM is marginary stable.

Hence we cannot determine the stability of the original nonlinear system.

#3. 범위 아님

#4. a. Larger, Periodic orbit such that is a solution of the nonlinear differential equations of motion.

b. 범위 아님